STREAM: RESEARCH-DERIVED FYP

ZKBlockchain

A Privacy-Preserving Blockchain Framework for Healthcare Data Sharing Using Post-Quantum Cryptography and Zero-Knowledge Proofs

Daniyal Qureshi (23I-0645) • Zia Ul Baqa (23I-0606) • Hannan Abid (23I-0713)
Supervisor: Dr. Noshina Tariq

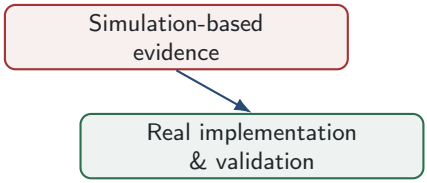
Fall 2026 • FYP-I Proposal Defense

Title & Problem Statement

Proposed FYP Title

A Privacy-Preserving Blockchain Framework for Healthcare Data Sharing Using Post-Quantum Cryptography and Zero-Knowledge Proofs

Short Title: ZKBlockchain
Team: Daniyal Qureshi (23I-0645)
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One-Sentence Problem Statement

Existing simulation-based blockchain architectures for Internet hospital systems demonstrate promising performance, but their security, privacy, scalability and reliability claims remain **insufficiently validated** under real cryptographic implementations, distributed network conditions and realistic physiological data.

Stakeholders & Current Data Workflow

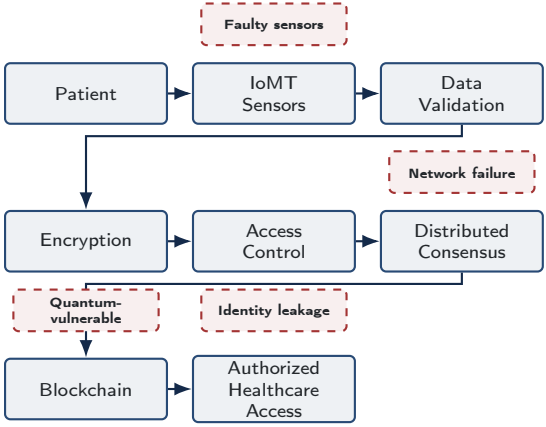
Who is affected?

- Patients – privacy & data integrity
- IoMT sensor devices – reliability of readings
- Healthcare professionals – authorized, timely access
- Hospital blockchain / IT infrastructure

Where can it fail?

- Quantum-vulnerable cryptography
- Identity / privacy leakage during authorization
- Unrealistic consensus assumptions
- Network delay, loss, partitions
- Faulty / spoofed sensor data

SDG 3 – Good Health & Well-Being



Existing Solution & Evidence-Based Gap

WHAT THE PAPER SHOWS → **WHAT IT DOES NOT YET ESTABLISH** → **WHAT OUR FYP WILL TEST**

Capability	Existing Paper	Gap	Proposed FYP
Post-quantum crypto	Kyber-inspired simulation	Not a certified PQC primitive	Real standardized PQC implementation
Privacy-preserving access	Hash commitment	Not a genuine ZK proof	Proper privacy-preserving proof mechanism
Consensus	Simulated Raft	No real distributed deployment	Real multi-node deployment
Network model	Synchronous / constant delay	Unrealistic WAN conditions	Delay, loss, partition, failure scenarios
PBFT comparison	Measured to 20 nodes + extrapolated	Unequal benchmark	Fair measured baseline at feasible scale
Sensor data	Synthetic	Limited real-world validity	Real / open physiological dataset
Security analysis	Restates primitive properties	Limited novel validation	Empirical security / threat testing

*Note: gaps are framed as **validation and implementation limitations**, not as invalidating the original paper's contribution.*

Research & Technical Basis

1. Post-Quantum Cryptography

- RSA is vulnerable to future quantum (Shor's algorithm) attacks
- Standardized PQC $>$ ad-hoc simulation for real security guarantees
- Candidate: *[INSERT SELECTED PQC ALGORITHM – e.g. CRYSTALS-Kyber / ML-KEM]*

2. Privacy-Preserving Access Control

- Hash commitment $\neq$ zero-knowledge proof (no soundness/ zero-knowledge guarantee)
- Identity/role disclosure risks patient and staff privacy
- Candidate: *[INSERT SELECTED ZK / PRIVACY MECHANISM]*

3. Distributed Consensus

- Raft (CFT) vs. PBFT / HotStuff / Tendermint (BFT)
- Crash fault tolerance vs. Byzantine fault tolerance
- Real network conditions materially affect throughput/latency

4. IoMT Data Reliability

- Sensor noise, missing values, spoofed/outlier readings
- Requires realistic physiological datasets for validation

Decision Table

Technical Question	Alternatives Considered	Selected Approach	Reason
Which PQC primitive?	RSA / ECC / lattice-based PQC	*[TBD]*	NIST-standardized, quantum-resistant
Which privacy mechanism?	RBAC / hash commitment / ZK proof	*[TBD]*	Verifiable without identity exposure
Which consensus?	Raft / PBFT / HotStuff	*[TBD]*	Byzantine resilience at feasible node count

Why This Is a Complex Computing Problem

This is not routine CRUD / system-integration work. It requires balancing conflicting engineering trade-offs:

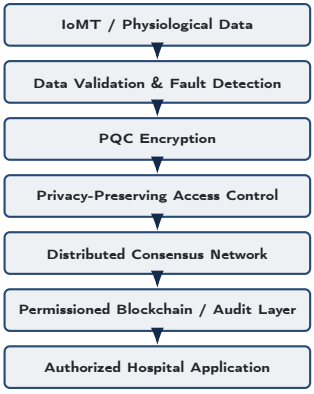
- **[T1] Security ↔ Performance** — stronger crypto/privacy increases computation & latency
- **[T2] Privacy ↔ Verifiability** — prove authorization without exposing identity/policy
- **[T3] Scalability ↔ Fault Tolerance** — more nodes change communication overhead & failure behavior
- **[T4] Realism ↔ Reproducibility** — real network/data conditions are harder to control
- **[T5] Reliability ↔ Complexity** — faulty sensor data cannot simply be trusted and stored

Seoul Accord Graduate Attributes Addressed

Complex problem analysis • Design under uncertainty • Trade-off evaluation • Integration of multiple technical domains (cryptography, distributed systems, networking, data science) • Experimentation and empirical validation

Requires research, experimentation, architecture decisions and engineering judgment — not routine implementation.

Proposed Contribution: System Architecture



In Scope

- Real PQC implementation
- Privacy-preserving authorization
- Distributed consensus prototype
- Realistic network simulation/emulation
- Physiological/IoMT dataset validation
- Measurable benchmarking

Out of Scope

- Real hospital deployment
- Clinical diagnosis
- Production-scale infrastructure
- Regulatory certification claims
- Clinical efficacy claims

Contribution Statement

We will implement and experimentally validate the security, privacy, distributed-system and sensor-reliability layers that the source study evaluated primarily through simulation, under controlled but substantially more realistic deployment conditions.

Initial Technical Spike — To Be Completed Before Final Proposal Approval

Risk Tested

Can real PQC key exchange operate together with multi-node distributed communication under non-ideal network conditions? This is the highest-risk assumption underlying the whole architecture.

Planned Experiment

- Implement PQC key encapsulation/decapsulation (candidate: ML-KEM/Kyber)
- Deploy ≥ 3 nodes on separate processes/containers
- Emulate latency/packet loss between nodes (e.g. `tc netem`)
- Measure round-trip crypto + propagation latency

Evidence (Planned)

- Console log of successful encapsulation/decapsulation
- Node-to-node message propagation trace
- Latency-vs-packet-loss benchmark chart

What We Will Learn / Design Decision

Whether the candidate PQC + consensus combination is computationally feasible at the node counts and network conditions targeted for FYP-1, or whether the scope must be adjusted (e.g. smaller cluster, alternate primitive).

No fabricated results are shown — this spike will be executed during early FYP-I

Evaluation Plan

Research Question	Baseline	Experiment	Metric	Success Criterion
Does real PQC give acceptable performance?	RSA-2048	Encryption workload	Latency, throughput, ciphertext size	[TO BE BASELINED DURING FYP-1]
Does the privacy mechanism reduce information exposure?	RBAC / hash commitment	Access requests	Proof time, leakage indicators, correctness	No unauthorized disclosure
Does consensus remain usable under realistic networks?	Baseline consensus	Delay / loss / partition scenarios	TPS, latency, finality, recovery	[TO BE BASELINED DURING FYP-1]
Does sensor correction generalize beyond synthetic data?	Raw / no correction	Real / open physiological data	Accuracy, MAE, fault detection	[TO BE BASELINED DURING FYP-1]

Non-Functional Criteria

- Recovery time after node failure
- Max. acceptable consensus latency
- CPU / memory overhead
- Network bandwidth / message overhead
- Cryptographic operation latency

Critical Failure Cases Tested

- Node crash • Network delay • Packet loss
- Network partition • Invalid authorization
- Replay attempt • Faulty/spoofed sensor reading
- Excessive sensor faults

Scope, Risks & Dependencies

Committed Scope

- **Real PQC key exchange implementation**
- **Privacy-preserving authorization prototype**
- **Multi-node distributed consensus deployment**
- **Realistic network emulation experiments**
- **IoMT fault detection & real dataset validation**

Optional Extensions

- **Full WAN (multi-region) deployment**
- **Second independent physiological dataset**
- **Formal / mechanized proof of the ZK protocol**

Key Risks → Mitigation / Fallback

- ZK integration too complex → **well-defined privacy-preserving proof prototype, benchmarked against original hash-commitment**
- Real physiological dataset unsuitable → **use a second open dataset, same evaluation protocol**
- Full WAN deployment infeasible → **controlled network emulation across multiple distributed nodes**
- PQC / distributed debugging complexity → **start with reduced node count, scale incrementally**

Dependencies: compute resources • open datasets • modern PQC/ZK libraries • containerized/network-emulation tooling • no real patient data used

Team Responsibilities & FYP-I / FYP-II Plan

Daniyal Qureshi
Post-quantum cryptography implementation & secure data storage

Zia Ul Baqa
Privacy-preserving / zero-knowledge access control & security evaluation

Hannan Abid
Distributed consensus, network realism, IoMT data pipeline & fault tolerance

Shared: system integration • testing • benchmarking • documentation • security review • final demo

FYP-I

Literature & gap validation → requirements → architecture → technical spike / POC → baseline implementation → dataset preparation → initial experiments

FYP-II

Full integration → realistic network experiments → security/privacy evaluation → sensor validation → performance benchmarking → final refinement → demonstration & thesis

FYP-I proves feasibility; FYP-II proves the contribution.